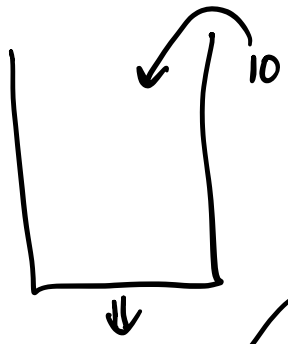


$x = 10$

name = "Akanksha".
↳ string

Variables → label
→ variable name.

Variable Name

→ A-Z, a-z, _, 0-9 → Valid characters

→ can't start with digit. → Are Case-Sensitive

- _age ✓ valid
- 12 age ✗ Invalid.

age = 20

To check type → type (age)
<class int>

Datatypes

Primitive
↓
Atomic
→ can't be disintegrated

Non-Primitive
↳ depend on other datatypes.

Dynamic Types and Literals

$x = 10$ → During run-time, this type is assessed.
↳ This is dynamic-typing.

- Literals: Numeric literals
  Boolean literals. <class 'bool'>

- A literal is a fixed value in the source code, such as numbers, strings or boolean values. Once defined, it remains constant during program execution.

Taking input

user_phone_number = input()

input() always stores value in string.
↓
Thus, Type Conversion is used.

int[input()]

9999 → processed as string
then passed to int.

Operators

→ **Arithmetic**

operation → operator
→ operand – on what value we want to do operation.

operands
2 + 3
↓
operator.

- + Add → In case of string → Concatenation.
- − Subtract
- * Multiply → works for string as well
- / divide → Returns float
- // Floor divison → Returns int
- % Remainder.
  modulo
- ** exponent {eg $2^3 \rightarrow 8$} → 2 ** 3 → 2^3.
  ↳ Error with string

Error with str.

Error with string

• For bool if we give $\begin{cases} +ve \\ -ve \end{cases}$ → bool = True
↳ if 0 → bool = False.

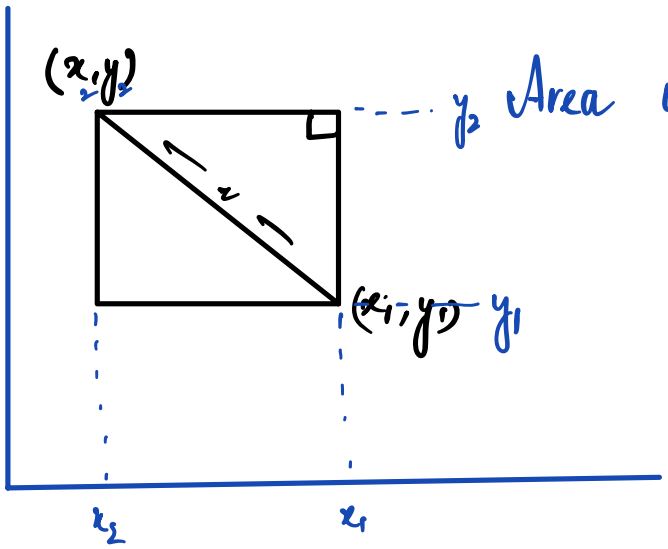
Comments

$\boxed{\text{or}}$ ''' ... '''

Helps write multiline comment.

Area of Rectangle

(If rectangle is parallel to axis)



Area of rectangle $= (x_2 - x_1) \times (y_2 - y_1)$